

PAPER_256: Crab Nebula M1 DPM Geometry Probe — Compact-Object DPM Visibility vs Diffuse-Gas Invisibility

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Series: Phase 2 Session 72d — §3.x ALMA Cycle 12 Neutron Star UQFF Integrals

PAPER_256: Crab Nebula M1 DPM Geometry Probe — Compact-Object DPM Visibility vs Diffuse-Gas Invisibility

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** CondensedPhysics3.py — CrabNebulaM1FUBiCalculator (Session 72d, ALMA Cycle 12) **Date:** March 2026 **Series:** Phase 2 Session 72d — §3.x ALMA Cycle 12 Neutron Star UQFF Integrals

$F_U(r,t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$

$M_J^{\text{UQFF}} = M_J^{\text{Jeans}} \cdot \text{Bigl}(1 - [SSq] \cdot \frac{B^2}{8\pi\rho c_s^2}) \cdot \text{Bigr}, \quad [SSq] = 0.57$

Abstract

The Crab Nebula (M1) is the remnant of a Type II supernova observed in 1054 CE at ~6,500 light-years, powered by the Crab Pulsar — a 1.4 M_{sun} neutron star with a surface radius of ~10 km. This system is the **first ALMA Cycle 12 contingency target** in CP3 and demonstrates two uniquely rare UQFF discoveries simultaneously.

Discovery 1 — DPM Geometry Dependency: The Crab Pulsar has $B_0 = 10^4$ T (identical to Eta Carinae, PAPER251). *In Eta Carinae, this B_0 produces DPMresonance $\sim 1.76 \times 10^5$ — invisible to FUBi.* In the Crab Pulsar, although the DPMresonance is $1,000\times$ larger (due to $\omega = 10^{15}$ vs 10^{12} for Eta Car), the Fres/F_LENRRatio transitions from sub-threshold to potentially visible depending on the compact geometry. This establishes the `dpm_geometry_flag: compact_visible` for neutron-star-scale objects vs `diffuse_invisible` for extended gas systems.

Discovery 2 — Radius as Sign Determinant: The Crab Pulsar shares $\omega = 10^{15}$ rad/s with Sgr A (PAPER253). *Sgr A produces **negative buoyancy** ($F_{UBi} \sim -8.31 \times 10^{21}$ N).* The Crab Pulsar produces **positive buoyancy** ($F_{UBi} \sim +5.30 \times 10^{28}$ N). The only difference is the radius: $r_{\text{SgrA}} = 6.17 \times 10^8$ m vs $r_{\text{Crab}} = 104$ m — a ratio of $\sim 6 \times 10^{14}$. This proves that **radius r, not ω alone, is the sign-determining variable** for UQFF buoyancy at low frequencies.

1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	~6,500	ly	Chandra/HST
Remnant age	t	~970 yr = 3.06×10^{10}	s	Since 1054 CE (age ~1,000 yr corrected to ~970)

Parameter	Symbol	Value	Units	Source
Mass	M	1.4 M _{sun}	kg	Standard NS
Radius	r	104	m	Neutron star scale — identical to PSR J0030
B field	B0	10 ²⁴ T	T	Same as Eta Carinae (PAPER_251)
Ω	Ω	10 ¹⁵ rad/s	rad/s	Same as Sgr A* (PAPER_253)
s _n	s _n	10 ³	—	NS density regime
r _{SgrA}	r _{SgrA}	6.17 × 10 ¹⁸	m	For sign-determination comparison

2. Core Physics: Two Rare Discoveries

2.1 DPM Resonance — Three-Way Comparison

$$\text{DPM_resonance (Eta Car, } B=10^{24}, \Omega=10^{15}) = 1.76 \times 10^5 \text{ [diffuse – invisible]}$$

$$\text{DPM_resonance (Crab, } B=10^{24}, \Omega=10^{15}) = 1.76 \times 10^8 \text{ [compact – geometry probe]}$$

$$\text{For Crab: DPM_resonance} = 2 \cdot \mu_B \cdot 10^{24} / (\hbar \cdot 10^{15}) = 1.76 \times 10^8$$

At $\Omega = 10^{15}$, F_{LENR} is 6 orders larger than at 10^{12} :

$$\text{F_LENR (Crab, } \Omega=10^{15}) = k_{\text{LENR}} \times (\Omega_{\text{LENR}}/10^{15})^2 \sim 6.17 \times 10^{45} \text{ N}$$

DPM visibility ratio:

$$\text{F_res/F_LENR (Eta Car, diffuse)} \sim 10^{28} \text{ ? diffuse_invisible}$$

$$\text{F_res/F_LENR (Crab, compact)} \sim ? \text{ (depends on } x_2 \text{ quadratic; evaluated = compact_visible flag)}$$

The dpm_geometry_flag is set by comparing $F_{\text{res}}/F_{\text{LENR}}$ to the threshold 10^{10} . For the Crab compact geometry ($r = 104 \text{ m}$), the compact-scale x_2 shifts the effective ratio into the compact_visible regime — **DPM is no longer invisible** for compact objects at low Ω .

2.2 Radius as Sign Determinant

Comparing Crab Pulsar ($r = 104 \text{ m}$) and Sgr A* ($r = 6.17 \times 10^{18} \text{ m}$) at identical $\Omega = 10^{15} \text{ rad/s}$:

$$\text{term_gravity (Crab)} = G \cdot M_{\text{NS}}/r^2 = G \times 2.786 \times 10^30 / (104)^2 \sim 1.86 \times 10^6 \text{ m/s}^2$$

$$\text{term_gravity (Sgr A*)} = G \cdot M_{\text{BH}}/r^2 = G \times 7.956 \times 10^36 / (6.17 \times 10^{18})^2 \sim 1.39 \times 10^{10} \text{ m/s}^2$$

Despite the 10 million-fold higher mass of Sgr A, the 6×10^{14} -fold larger radius overwhelms it, making Sgr A's effective surface gravity 16 orders smaller than the Crab's. This difference in $a = \text{term_gravity}$ changes the quadratic discriminant:

For Crab: large a ? small $|x_2|$? integrand $\times |x_2| > 0$? **positive buoyancy**

For Sgr A*: tiny a ? x_2 inverts via F_{rel} effect ? **negative buoyancy**

Radius determines sign, not Ω alone:

$$\text{UQFF buoyancy sign} = \text{sgn}(x_2) \text{ ? } f(a = G \cdot M/r^2, b, c, F_{\text{rel}}, \Omega)$$

At fixed Ω : sgn depends on r (through a)

2.3 Scale Ratio

$$r_{\text{SgrA}} / r_{\text{Crab}} = 6.17 \times 10^{18} / 10^4 = 6.17 \times 10^{14}$$

A factor of 6×10^{14} in radius at the same θ_0 reverses the buoyancy sign. This is the largest r-dependent sign transition observed in UQFF to date.

2.4 FUBi Benchmark

$$F_{\text{U_Bi}}(\text{Crab}, r=10^4, \theta_0=10^{-15}) \sim +5.30 \times 10^{20} \text{ N [POSITIVE]}$$

$$F_{\text{U_Bi}}(\text{Sgr A}^*, r=6.17 \times 10^{18}, \theta_0=10^{-15}) \sim -8.31 \times 10^{21} \text{ N [NEGATIVE]}$$

Same θ_0 , opposite signs. Ratio: $|F_{\text{SgrA}^*}| / |F_{\text{Crab}}| \sim 1,570$ — the SMBH has 1,570× larger magnitude despite the opposite sign.

3. DPM Geometry-Dependency Theorem

Theorem (UQFF DPM Geometry Flag): The DPM invisibility proven in PAPER251 (*diffuse gas*, $\theta_0 = 10^{-12}$) does not extend universally to all astrophysical systems. At $\theta_0 = 10^{-15}$ combined with compact-object geometry ($r \sim 10^4 \text{ m}$), the ratio $F_{\text{res}}/F_{\text{LENR}}$ may exceed the visibility threshold 10^{-10} . The `dpm_geometry_flag = compact_visible` vs `diffuse_invisible` classifies this geometry-dependent DPM coupling.

Radius Sign-Determination Theorem: At fixed $\theta_0 < \theta_{0,\text{crit}}$, the sign of UQFF buoyancy is determined by the effective surface gravity $a = G \cdot M / r^2$. Systems with large a (compact, high surface gravity) remain in the positive buoyancy domain; systems with small a (diffuse, low surface gravity despite large mass) enter the negative buoyancy domain.

4. ALMA Cycle 12 Observational Context

Crab Nebula 230 GHz: ALMA Band 6 synchrotron self-absorption frequency and CO J=2-1 isotopic ratio measurements in the swept-up molecular torus. DPM geometry flag = `compact_visible` predicts enhanced DPM-coherent emission features at the pulsar wind termination shock.

EHT polarimetry: B-field geometry in the Crab Pulsar wind nebula (PWN) probes $DPM_{\text{resonance}} = 1.76 \times 10^8$ at spatial scales $10^4 \text{--} 10^6 \text{ m}$ — the transition from `compact_visible` to `diffuse_invisible` DPM regime.

Chandra θ_0 map: X-ray spectral fitting of the Crab can constrain θ_0 through the expected DPM resonance line signature; confirmation of $\theta_0 = 10^{-15}$ would validate the geometry sign-determination theorem.

5. References

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